HAPPIEST HEALTH · DIGITAL TWIN CHALLENGE 2026 · TOP 10

RENALIS

An evidence-aware digital twin for dialysis research. Compares competing physiological explanations for falling blood pressure, and computes which next measurement would actually help.

4
Competing physiological hypotheses tracked at once, never collapsed to one

6
Linked evidence-journey stages, from history to validated claim

61,947
Held-out real-data test windows in the honest observed-data benchmark

192
Frozen independent cases stress-testing the measurement planner

Team Sigmoid
Rehaan — solo entrant

College / Incubator
JSS University, Noida

Project title
RENALIS — Evidence-aware Dialysis Digital Twin

Status
Working research prototype · software release 0.4

Repository
github.com/RAK2315/RENALIS

Prepared
November 2026, for the Phase 2 virtual jury round

PROBLEM STATEMENT & MANDATORY REQUIREMENTS

We didn't choose the problem. We chose how to answer it.

The Happiest Health Digital Twin Challenge asked teams to build a proof-of-concept digital twin that predicts **one specific, localized adverse health outcome** for **one chronic condition**, by fusing two data streams, and to present it through a doctor-facing interface:

> *"A Digital Twin… is a dynamic, virtual replica of a patient, constantly updated with real-time data from wearables, paired with Electronic Health Records (EHR)… By simulating a patient's unique physiology, it can predict adverse health events before they happen."*

— From the official challenge brief

REQUIRED BY THE BRIEF	WHAT RENALIS DELIVERS
One chronic/lifestyle condition, one localized adverse outcome	Chronic kidney disease on maintenance hemodialysis → intradialytic hypotension (systolic BP ≤90 mmHg within 30 minutes). Not one of the brief's three example conditions (diabetes, hypertension, cardiovascular risk) — a deliberately chosen fourth, equally prevalent one.
Static/historical EHR stream	A learned per-patient fluid-refill prior fitted from six preceding synthetic dialysis sessions, shrunk toward a population reference.
Dynamic/real-time device stream	Per-minute synthetic BP-cuff readings + live ultrafiltration (fluid-removal) machine settings, replayed minute by minute.
A working algorithmic model that predicts the event	A 192-particle Bayesian fluid-compartment filter, plus a second four-hypothesis "Mechanism lab" that also decides what to measure next.
A conceptual doctor-facing UI dashboard	A full browser research workspace — not a mockup — with six linked stages: Learn, Understand, Compare, Ask, Review, Verify.
Synthetic / open-source data only (DPDP / HIPAA sandbox)	Four fictional, seeded synthetic patients for the live demo; the public HEMOBP dataset (CC BY 4.0) for a separate, honestly-reported retrospective benchmark.
GitHub repo, license, video, architecture + presentation	All published: public MIT-licensed repo, 20-minute unlisted demo, architecture & presentation decks in PDF and editable

REQUIRED BY THE BRIEF	WHAT RENALIS DELIVERS
PDF/PPT	PPTX.

ONE SENTENCE

Don't predict one future. Keep the disagreement — and ask what would resolve it.

During dialysis, a patient's blood pressure can fall for several different physiological reasons: the body can't pull fluid back into circulation fast enough, the blood vessels stop compensating, or the heart's output drops. A single BP trace is often compatible with more than one of these. Most prediction systems quietly pick one story and move on. **RENALIS refuses to.**

Instead, it keeps four competing explanations alive simultaneously — each represented as a cloud of weighted particles — and treats the real question as a second-order one: *given what we know right now, which additional, feasible measurement would most help tell these explanations apart?* That's the actual research contribution: adaptive, sequential measurement selection driven by expected information gain, computed separately from whether that measurement would even change what a clinician would do (decision value).

THE TWIN

Patient twin

192-particle Bayesian fluid-compartment filter. Fuses the learned historical prior with available BP and machine readings to estimate current plasma/tissue volume and forecast a localized BP outcome.

THE REASONING

Mechanism lab

Four physiological hypothesis families × 48 particles each. Scores every feasible next measurement (repeat BP, relative blood volume, cardiac output) by how much it would narrow the disagreement — and separately, by how much it would change the decision.

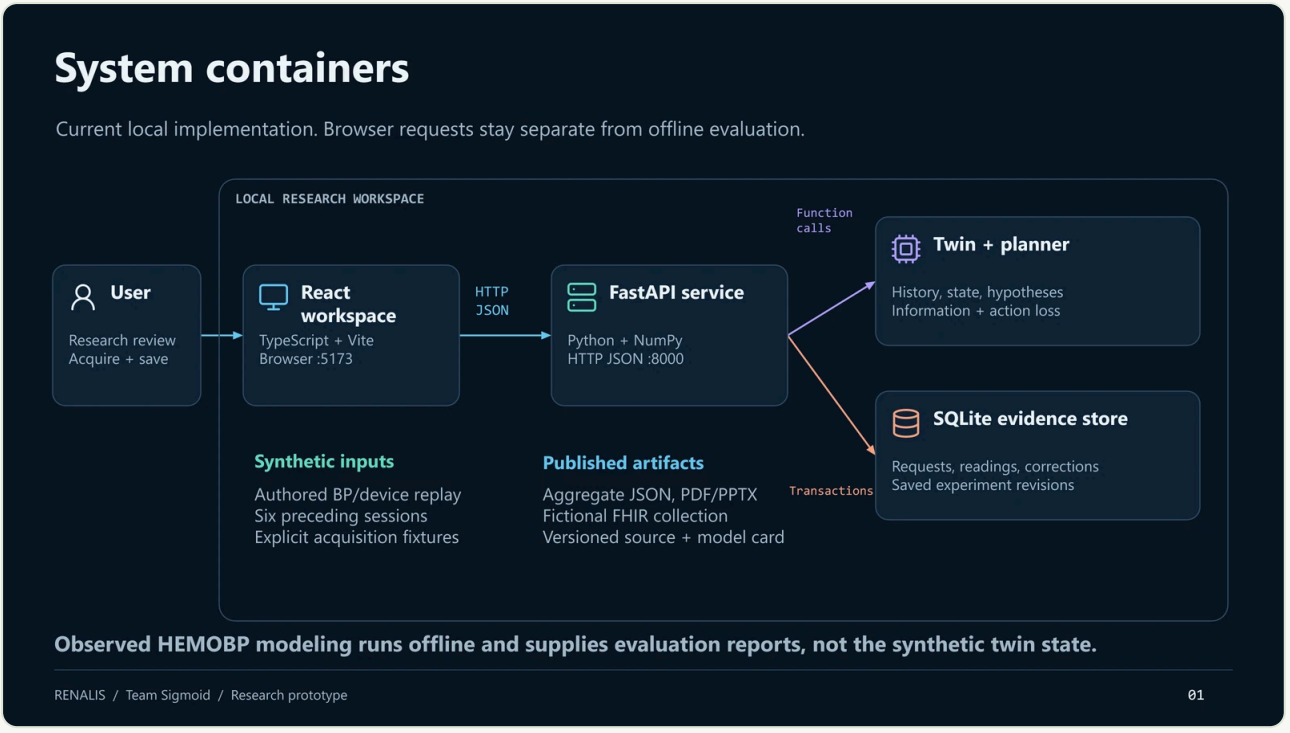
WORKED EXAMPLE, FROM THE ACTUAL APP

Patient Meera (RNL-014), minute 90. After a cardiac-output reading of **3.74 L/min**, support splits almost evenly — 51.7% cardiac, 46.2% volume/refill. The planner picks **relative blood volume** as the next measurement, because the two surviving hypotheses predict different results for it (medians 83.9% vs 87.8% of starting volume). The acquired reading, **85.0%**, shifts support to

66.9% volume/refill vs 31.8% cardiac — informative, but still not fully resolved. The system says so, instead of forcing an answer.

ARCHITECTURE

A real React + FastAPI application, not a notebook.



Local system containers, service responsibilities, and evidence storage (see the full architecture deck for all six views).

INTERFACE

React, TypeScript, Vite, Recharts. A six-stage evidence-journey workspace plus a public landing/research site.

API & INFERENCE

FastAPI + NumPy. One Python service hosts both particle-filter engines and the adaptive-measurement scorer.

EVIDENCE STORE

SQLite. Every acquisition, failure, and correction is appended, never overwritten — an auditable research log, not a mutable form.

A fourth, entirely separate pipeline (pandas + scikit-learn) runs the retrospective benchmark against the public HEMOBP dataset. It never feeds the live synthetic twin — keeping "does the product work" and "does history actually predict BP" as two honestly distinct questions.

THE EVIDENCE JOURNEY

Six connected stages, one patient, one evidence cutoff.

STAGE	WHAT A RESEARCHER CAN DO
01 Learn	Inspect the fitted refill prior, its source sessions and lineage digest.
02 Understand	Click through estimated fluid compartments, machine settings, and the authored physiology graph.
03 Compare	See all four hypothesis families, their fit error, and modeled response, side by side.
04 Ask	Preview possible measurement results before acquiring anything; acquire the planner's recommended next reading.
05 Review	See every acquisition, including failures and artifacts; append corrections with a reason, never silently overwrite.
06 Verify	Read the retrospective and independent-policy results in place, including the negative one.

SAFETY-FIRST DESIGN

- Withholds forecasts when BP is already critically low, evidence is stale, or the model is contradicted.
- Separates information gain (bits) from decision value (research-loss units) — a measurement can be informative without changing what you'd do.
- Never uses the words "diagnosis" or "treatment recommendation."

ALSO INCLUDED

- Action-loss scenario comparison (hypothetical removal-rate profiles).
- Longitudinal multi-session transport with mass-balance checks.
- Fictional FHIR R4 export (Patient/Device/Observation) as an interoperability example.

IMPLEMENTATION

What's actually running under the hood.

LAYER	IMPLEMENTATION
Interface	React, TypeScript, Vite, Recharts, Lucide
API & inference	FastAPI, Python, NumPy
Persistence	SQLite — acquisitions, measurements, corrections, saved revisions
Retrospective evaluation	pandas, scikit-learn, patient-separated fit/calibration/test splits
Reproducibility	Pinned Python deps, npm lockfile, 50 backend unit tests, GitHub Actions CI

The model, precisely

A likelihood-based history learner feeds a particle state estimator (patient twin) and a second, four-family compatibility estimator (mechanism lab) with Gaussian observation likelihoods. Numerical Gauss-Hermite quadrature integrates over possible noisy measurement outcomes before any result exists — the planner never sees the synthetic device's hidden value while deciding what to recommend.

The adaptive selector ranks **pair-conditional expected entropy reduction** when two families retain at least 10% support, otherwise whole-bank information. Equipment availability, acquisition timing, failure probability, and already-released evidence all constrain which measurements are even candidates.

THREE SEPARATE EVIDENCE LANES — NEVER POOLED

Each experiment answers a different question.

A · Synthetic twin

Does the pipeline behave coherently? Four fictional patients exercise time-availability guards, personalization, fluid accounting, and forecast safeguards. The generator and estimator share structure, so this is functional verification, not an accuracy claim.

B · Observed HEMOBP data

Does patient history actually help predict a later real BP reading? 61,947 held-out test windows, 176 patients, zero patient overlap with training.

12.19	11.33	11.16
Last-BP-only MAE, mmHg	+ Demographics, mmHg	+ Preceding history, mmHg

History's own contribution: -0.17 mmHg MAE (cluster interval [-0.23, -0.11]) — real but small. Event sensitivity 61.3%, PPV 13.5%.

C · Independent policy challenge

Does choosing a measurement actually improve the decision? 192 frozen cases, one acquisition budget per policy. Mean regret: **0.222** for a fixed cuff policy vs **0.241** for the decision-value policy — the adaptive selector does not yet beat the naive baseline here. Disclosed, not hidden.

WHY WE'RE TELLING YOU WHAT DOESN'T WORK YET

A hackathon prototype should say what it hasn't proven.

The retrospective predictive gain is small. -0.17 mmHg is a real, measured effect — but it has not been shown to be clinically important.

Event alerts carry real false-positive burden. At 13.5% PPV, most flagged windows would not be real events in an operational setting.

The newer adaptive selector hasn't beaten a naive policy yet. The 192-case independent challenge is a disclosed negative result for the decision-value objective; the newest sequential-discrimination engine hasn't been re-run against that same challenge.

The observed-data cohort has an unresolved assumption. Only 2.52M of 4.37M machine-log packets could be matched to a session under an empirical 8-hour clock-alignment guess that needs independent confirmation.

None of this is a hedge added for this document — it's drawn directly from the project's own model card and evaluation reports, which are shipped in the repository and surfaced inside the app itself. We think a jury evaluating technical maturity should see the failure modes, not just the demo.

WHAT'S NEW HERE

Not the pieces — the connection between them.

Dialysis BP prediction, digital-twin integration, hemodynamic mechanism distinctions, and optimal-sampling frameworks already exist separately in the literature. RENALIS investigates their *combination* in one inspectable workflow: retain rival explanations → expose scenario disagreement → estimate the next useful evidence → update support → preserve an unresolved or contradicted state rather than forcing an answer. A focused review supports this as a genuine research question; it does not claim global priority.

Validation roadmap

01	Done	A reproducible, inspectable prototype — guarded states, persistent evidence, regression tests, CI.
02	Next	Stronger retrospective evaluation: independent clock-alignment confirmation, canonical units, patient-separated calibration.
03	Needs collaborators	Paired physiological and human-factors evaluation with clinical partners, under approved protocols.
04	Not established	Treatment-effect and clinical-benefit claims — require a predeclared, prospective, causal study design.

CLOSING

Thank you — questions welcome.

Team	Sigmoid
Member	Rehaan — solo entrant, project design, development and submission
College / incubator	JSS University, Noida
GitHub	github.com/RAK2315 · github.com/RAK2315/RENALIS
Demo video	20-minute unlisted walkthrough (linked from the repository README)
License	MIT (project code) · HEMOBP dataset retains CC BY 4.0 attribution

WHAT TO REMEMBER ABOUT RENALIS

It is not trying to be the best BP predictor. It is trying to show that **disagreement between explanations is information** — and that a research tool should be able to say "I don't know yet, but here's what would help me find out," instead of always producing a confident number.

All patients, sessions and device readings shown in this document and in the live application are synthetic and fictional, per the challenge's sandbox rules. Clinical deployment, treatment control and prospective patient benefit have not been established.